

Assignment 1

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Part 1 # foundational practice problems

Question: 1:

Take a number N and print numbers from N down to 1?

Algorithm:

1. Start
2. Input number
3. Set $i = 1$
4. ~~By~~ While $i \geq 1$ then
 print i
5. Otherwise decrease
 i by 1
6. repeat until $i < 1$
7. Stop.

pseudocode #

Start

Input Number

$i \leftarrow N$

While $i \geq 1$

 print i

$i \leftarrow i - 1$

End While

End

Question 2*

Input a year and determine whether it is a leap year. (Rule: divisible by 4 AND (not divisible by 100 OR divisible by 400)?)

Algorithm*

1. Start
2. Input year
3. Check whether the year is divisible by 4 AND (not divisible by 100 OR divisible by 400)
4. If the condition is true, then print "Leap year".
5. Otherwise print "not a leap year".
6. Stop.

Pseudocode*

Start

Input year

If $\text{year} \% 4 == 0$ AND

$\text{year} \% 100 \neq 0$ OR $\text{year} \% 400 = 0$ then

Print Leap year

Else

Print not a leap year

End If

Question # 3:

Take a number, count how many digits it has, and state whether it is a single-digit, double digit, or triple digit?

Algorithm #

1. Start
2. Input number
3. Set $i = 0$
4. While $N > 0$
 divide N by 10
 Increase i by 1
5. check value of i
6. If $i = 1$, print
 Single digit.
7. Else if $i = 2$, print
 double digit
8. Else if $i = 3$, print
 triple digit
9. display number of
 digits
10. stop.

Pseudocode #

Start

Input number
 $i \leftarrow 0$

While $N > 0$ DO
 $N \leftarrow N/10$
 $i \leftarrow i + 1$

END While

Number of digits = i

If $i == 1$ then
 print single digit
Else if $i == 2$ then
 print double digit

Else
 print triple digit or more
END If

END

Question # 4:

Print the multiplication tables from 1 to 10, but skip the table of any number that is multiple of 3?

Algorithm #

1. Start

2. Set $i = 1$

3. Repeat While $i \leq 10$

4. Check if it is Multiple of 3

5. If yes, skip the table.

6. Otherwise, set $j = 1$

7 - Repeat While $j \leq 10$

calculate $i \times j$

display result

Increase j by 1

8 - Increase i by 1

9 - Repeat until $i > 10$

10 - stop

pseudocode

Start

$i \leftarrow 1$

While $i \leq 10$ DO

If $i \% 3 == 0$, then

$i \leftarrow i + 1$

continue

End if

$j \leftarrow 1$

While $j \leq 10$ DO

$i * j$

$j \leftarrow j + 1$

End While

$i \leftarrow i + 1$

End While

End

Question 5+

Take a number, repeatedly sum its digits until the result is a single digit, and display each intermediate result. for example, if number is 123, the sum of its digits is 6?

Algorithm+

1. Start

2. Input number N

3. While $N \geq 10$

Set $Sum = 0$

While $N > 0$

find last digit using $N \% 10$

Add digit to sum

Remove last digit using $N / 10$

Set $N = Sum$

Display N

4. Stop

pseudocode

start

Input N

While $N \neq 0$

Sum $\leftarrow 0$

While $N > 0$

digit $\leftarrow N \% 10$

Sum $\leftarrow \text{sum} + \text{digit}$

$N \leftarrow N / 10$

END While

$N \leftarrow \text{sum}$

output N

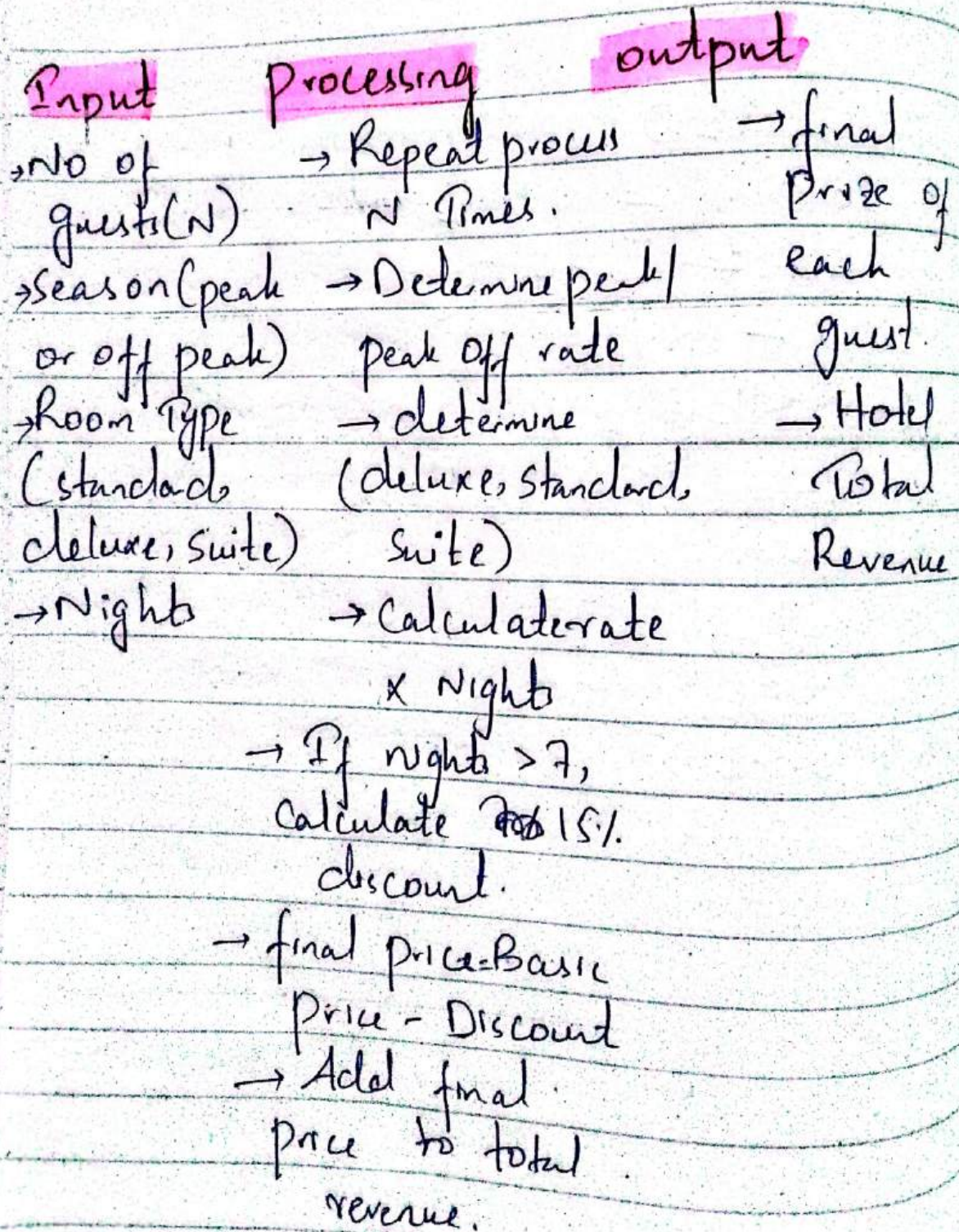
End While

END

part B#

Question 1#

IPO chart:



PAC chart

Input

- Number of guests N
- Season (peak / off-peak)
- Room Type
- Number of nights

Processing

- select rate according to season and room type
- calculate $\text{rate} \times \text{nights}$
- check if $\text{nights} > 7$
- Apply discount 15% if $\text{nights} > 7$.

Output

- Final prize of each guest
- Hotel Total Revenue

Solution Alternatives

- Season (peak / off-peak)
- Room Type: standard / deluxe / suite
- Stay duration: If more than 7 nights 15% discount
- If 7 nights or less than 7 no discount.

Algorithm

1. Start
2. Input number of guest N
3. Set Hotel total = 0
4. Repeat the following steps for each guest from 1 to N .

Input season

Input room type

Input nights

5. If season is peak

If room type is standard, set rate = 5000
Else If room type is deluxe, set rate = 8000
Else if room type is suite, set rate = 12000

6. Otherwise, if season is peak off

If room type is standard, set rate = 3000
Else if room type is deluxe, set rate = 5000
Else if room type is suite, set rate = 8000

7. Calculate Basic price = rate $\times$ nights

8. If nights > 7 ;

discount = Basic price $\times 15 / 100$

9. otherwise

discount = 0

10. calculate

final price = Basic price - discount

11. Print guest's final price
12. Add it to total

Hotel total = Hotel total + final price

13. After all N guests are processed, print Hotel total

14. End.

Pseudocode

Start

Input N

Hotel total $\leftarrow 0$

For guest $\leftarrow 1$ to N

Input season

Input room Type

Input nights

If season = "peak" Then

If Room Type = standard then

Rate $\leftarrow 5000$

Else If room type = deluxe then

Rate $\leftarrow 8000$

Else if room type = suite then
rate $\leftarrow$ 12000
END If

Else If season is "off-peak" then

If room type = standard then
rate $\leftarrow$ 3000

Else if room type = deluxe then
rate $\leftarrow$ 5000

Else if room type = suite then
rate $\leftarrow$ 8000

END If

Basic price $\leftarrow$ rate $\times$ nights

If nights $>$ 7 then
Discount $\leftarrow$ Basic price $\times$ 0.15

Else

Discount $\leftarrow$ 0

END If

Final price $\leftarrow$ Basic price $-$ discount

print final price

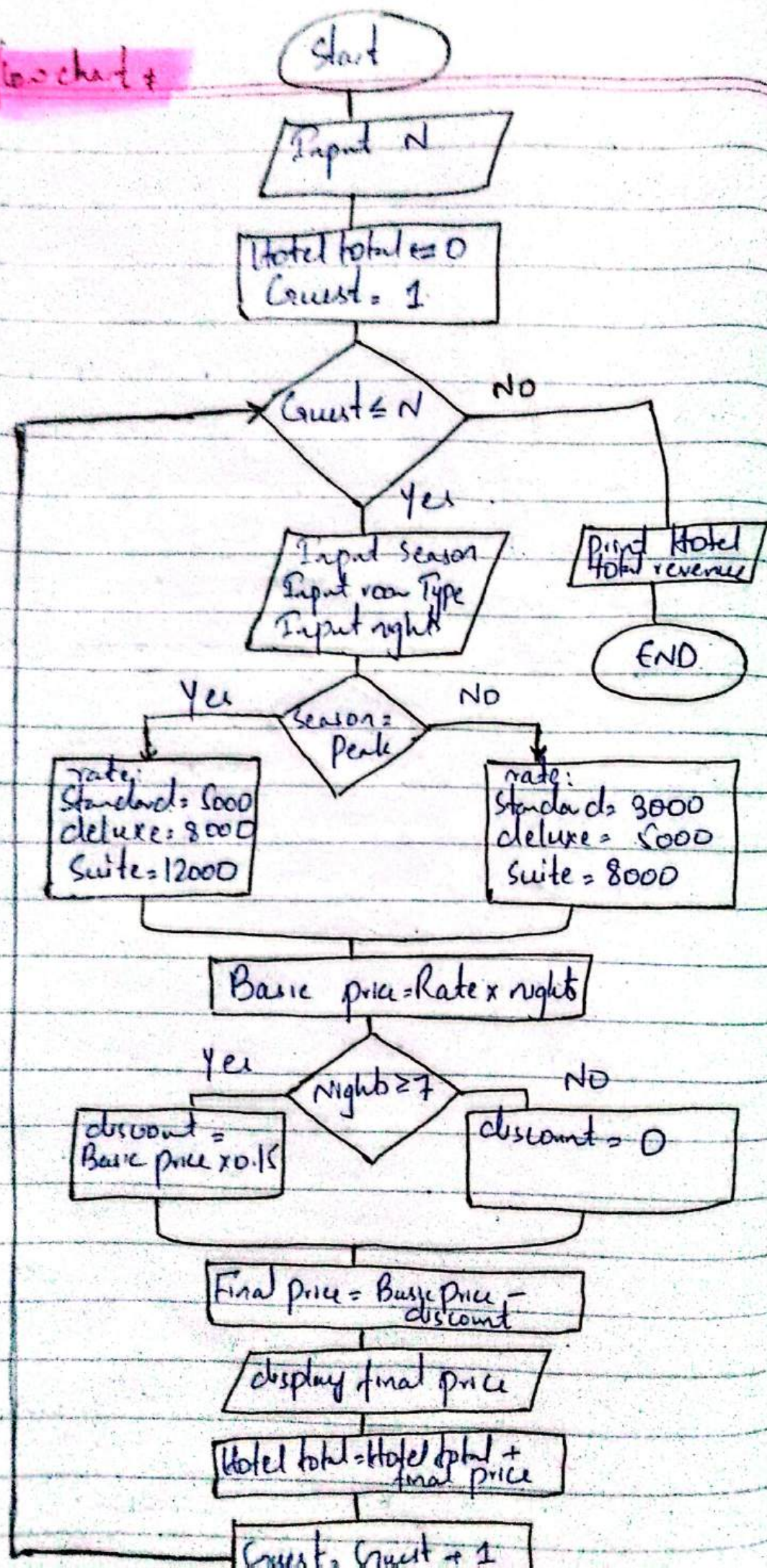
Hotel Total ← Hotel total + final price

END for

print Hotel Total

END

flowchart



Question 2#

IPO #

Input

- Number of request 'N'
- List of floor request
- Current floor (starts at 0)

processing

- compare each with current ~~request~~ floor.
- * If request > current → moving up

Output

- Moving up / Moving down doors opening for each request

- * If request < current → moving down

- * If request = current → doors opening

- update current floor to requested floor

- final current floor after all requests

PAC chart

Problem

Process a list of floor request for an elevator.

decide direction for each request

update current floor for each stop.

Alternatives

→ Starts at floor 0

→ If request > current
up

→ If request < current
down

→ If request = current
door opening

→ set current floor =
requested floor.

Consequences

Initial current floor is zero.

Correct movement message is printed.

Next comparison uses updated floor.

Alternatives

→ use a loop for n requests.

→ use if/else if/else for comparison

→ Initialize current floor = 0

Algorithm

1. Start
2. Input Number
3. Input List of requests
4. Set current floor = 0
5. for $i = 1$ to N , do steps 6-10
6. Input request
7. If request $>$ current, print 'move up'.
8. If request $<$ current, print 'move down'.
9. If request = current, print 'door opening'.
10. Set current floor = request
11. End for
12. print 'final floor' and 'current floor'.
13. End

Pseudocode

Start

Input N

Current floor $\leftarrow 0$

for $i \leftarrow 1$ to N

Input request

If request $>$ current floor then
print move up

Else If request $<$ current floor then
print move down

Else

print doors opening

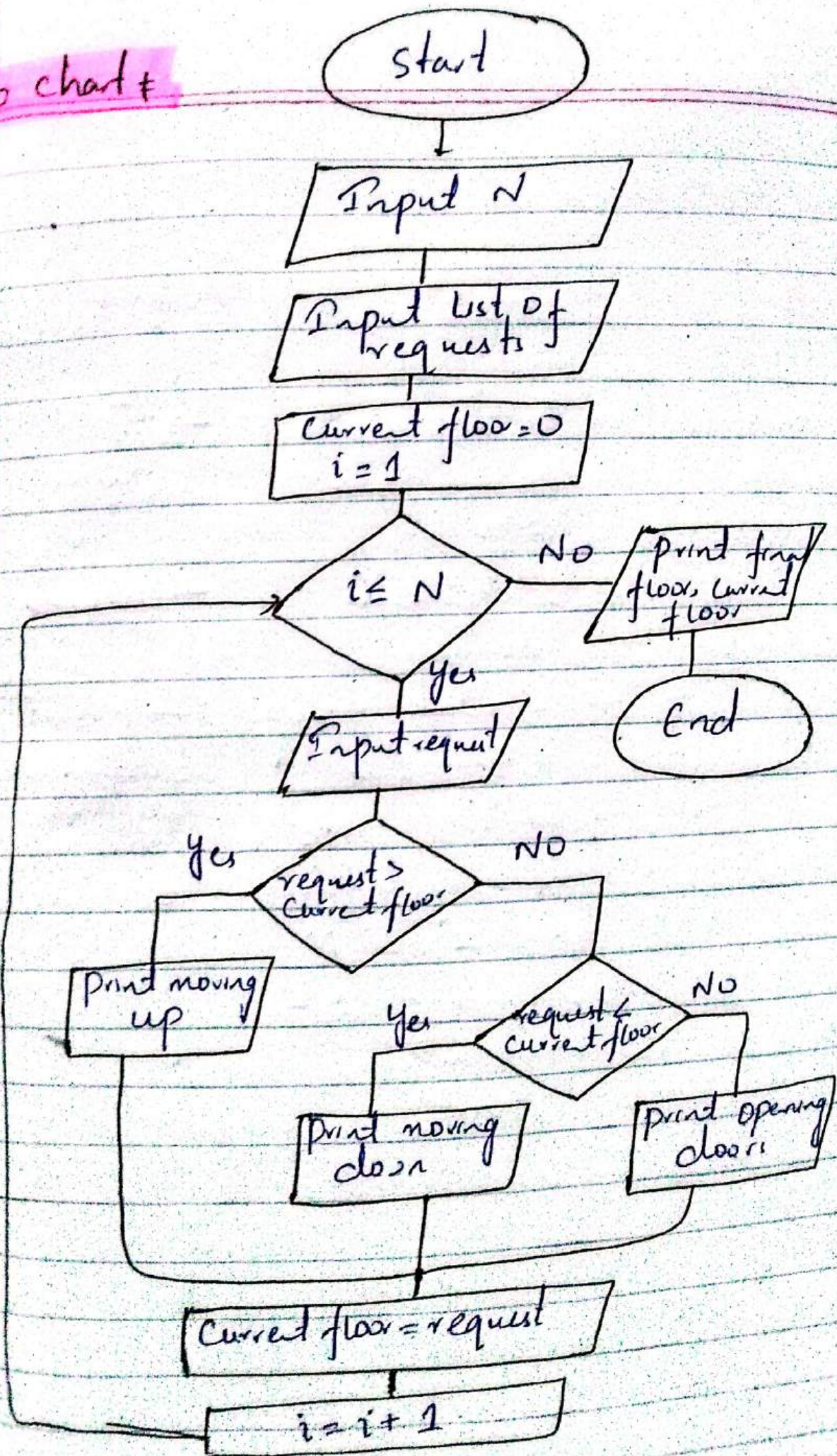
End if

Current floor $\leftarrow$ request

End for

print final floor : Current floor

flow chart #



Question 3#

Ipo chart

Input

Process

Output

Number of
Students
N.

→ Take marks
for each
Student

→ Total
marks

S Subject
marks

→ Add the S
marks

→ average

→ calculate average
 $= \text{Sum} / S$

→ Result:
distinction

→ check average
→ check if any
mark < 33

Pass/fail

→ If any mark < 33 , override
result.

PAC chart

Input

N students

S Subject marks

Processing

→ Process N students using outer loop

→ process S marks using inner loop

→ Sum all S marks

→ Average = Sum / S

→ Average ≥ 80 .
distinction

→ Average ≥ 60 & < 80
pass

→ Average < 60
fail

→ If any mark < 33
Subject deficiency

Output

→ Student's total

→ Student average

→ Student result

Solution Alternatives

→ use outer loop for Students

→ use inner loop for S subjects

→ use If / Else If / Else for average classification

→ use a variable such as Subject deficiency to

remember if marks < 33 .

→ If Subject deficiency = true, override result to fail.

Algorithm

1. Start

2. Input number of students N

3. Repeat for each student from 1 to N
Set $Sum = 0$

Set subject deficiency = false

4. Repeat 5 times for ~~sub~~ student's Subject

Input Subject mark

Add mark to Sum

If mark < 33 , set subject deficiency = true

5. Calculate:

$$\text{average} = \text{Sum} / 5$$

6. If average ≥ 80 , set result to distinction

7. Else if average ≥ 60 , set result to pass

8. Else set result to fail

9. If subject deficiency = true, change result to fail - subject deficiency

10. Print total average & result

11. Repeat for next student

12. End.

Pseudocode #

Start

Input N

for student $\leftarrow 1$ to N

Sum $\leftarrow 0$

Subject deficiency $\leftarrow 1$ to 5
Input mark

Sum $\leftarrow$ Sum + mark

If mark < 33 Then

Subject deficiency $\leftarrow$ Print

End If

End for

average $\leftarrow$ sum / 5

If average ≥ 80 Then

result $\leftarrow$ distinction

Else If average ≥ 60 Then

result $\leftarrow$ pass

Else

result ← fail

End If

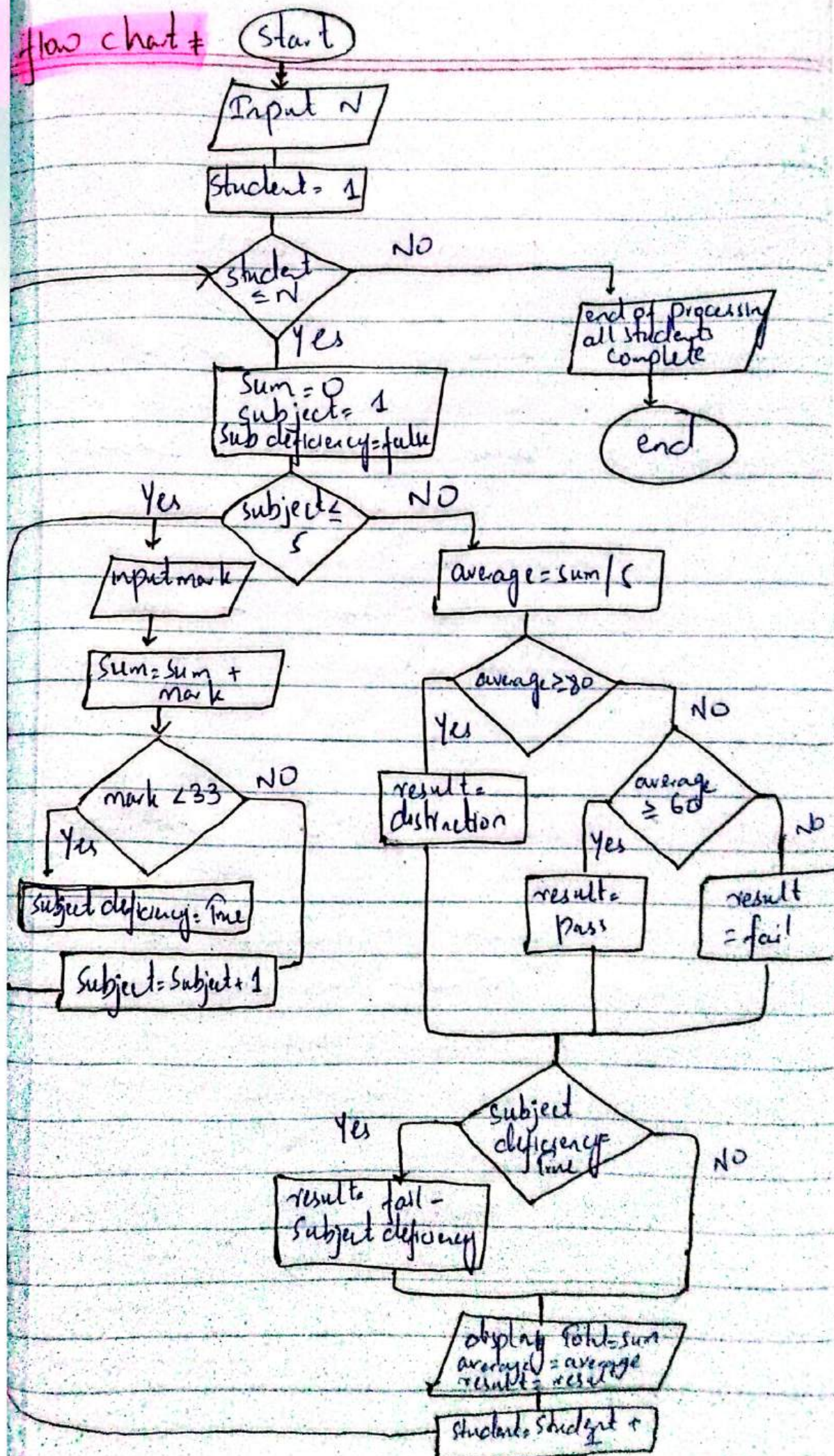
If subject deficiency = True Then
result ← fail - subject deficiency
End if

display Total = sum
display average, average
display result = result

End for

End

flow chart #



Question # 4:

IPO #

Input

process

output

Quantity q

Price per item p

Discount % d

Tax % t

$$s = q \times p$$

$$a = s - (s \times d) / 100$$

$$\text{bill} = a + (a \times t) / 100$$

validate all inputs

store calculation details

→ Subtotal

→ dis count

Amount

→ final Bill

→ error

message / Bill

PAC #

Input

process

output

Alternative

Quantity q

Price per item p

discount % d

Tax % t

→ validate

Input values

→ calculates =

$$q \times p$$

→ calculate $a =$

$$s - (s \times d) / 100$$

→ calculate bill =

$$a + (a \times t) / 100$$

→ subtotal

→ discounted

amount

→ final

Bill

→ shopping

bill

* If input

is invalid

→ show

error &

stop

* If valid

→ continue

calculation

→ apply discount

& tax

Algorithm

1. Start
2. Input quantity q
3. Input price per item p
4. Input discount percentage d .
5. Input tax percentage t
6. validate all entered values
7. If any value is Invalid:
 - display "Invalid Input"
 - Terminate the program
8. calculate sub-total $s = q \times p$
9. calculate discounted amount $a = s - \frac{(s \times d)}{100}$
10. Calculate final bill: $bill = a + \frac{(a \times t)}{100}$
11. Store calculation details
12. Generate & display the bill
13. End

Pseudocode

Start

Input q

Input p

Input d

Input t

If $q \leq 0$ OR $p < 0$ OR $d < 0$ OR $d > 100$
OR $t < 0$ OR $t > 100$ then

print "Invalid Input"

Stop

End If

$$s = q * p$$

$$a = s - (s * d) / 100$$

$$\text{bill} = a + (a * t) / 100$$

Store $q, p, d, t, s, a, \text{bill}$

Display "Shopping Bill"

Display "Quantity" = q

Display "Price" = p

Display "Subtotal" = s

Display "Discount" = d , "%" "

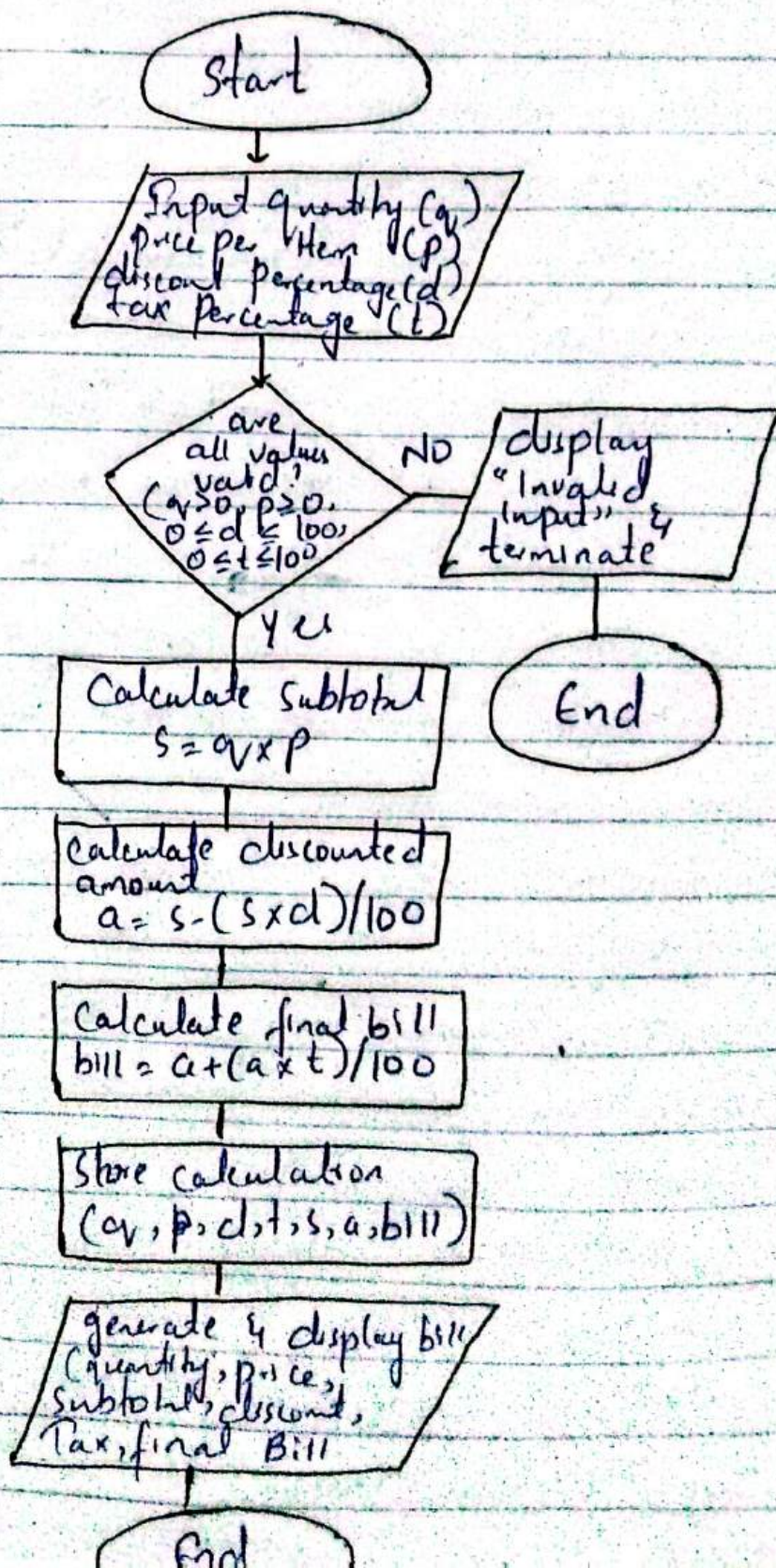
Display "discounted amount" = a

display "tax" = t , "%" "

display "final bill" = $bill$

END

Flow chart#



Question + S:

PPO +

Input

Number of vehicles (N)
vehicle Type (C, B, V)
user category (F/S/G)
permit Y/N
Emergency Y/N

Process

→ validate vehicle Type
→ validate user Category
→ validate permit Y/N
→ Ask emergency status if permit = N
→ check eligibility
→ check available Capacity

Output

→ Assigned Zone
→ Remaining Capacity
→ Rejection reason
→ parked cars count
→ parked bikes count
→ parked vans count
→ Rejected vehicle count
→ assign suitable zone
→ update zone capacity
→ Repeat for every vehicle.

PAC 1

Input

- Number of vehicles
- vehicle type: V, B, C
- user category: F, S, G
- parking permit: Y, N
- Emergency vehicle: Y, N
- Only when permit is (invalid)

Process

- validate entered Information
- check whether vehicle has valid Permit
- check emergency status if necessary
- determine appropriate Parking Zone

Alternative

- Invalid Input → ask again
- valid permit + space available → assign zone
- preferred zone
- full → check applicable alternative zone
- Repeat for all vehicles

No suitable space/zone

→ reject vehicle

No permit + emergency

Y → allow entry &

assign appropriate zone.

No permit + emergency

N → reject

Output

- Assigned Zone
- Remaining Capacity
- Rejection Reason
- Total parked cars
- Total parked bikes
- Total parked vans
- Total rejected vehicles

Algorithm

1. Start

2. Input number of vehicle N

3. Set Zone A capacity = 20, B = 40, C = 15

4. Set all counters to 0

5. Repeat for each vehicle from 1 to N

a. Input vehicle type

b. If vehicle type is invalid, ask again

c. Input user category

d. If category is invalid, ask again

e. Input parking permit

f. If permit is invalid, ask again

g. If permit is N, Input emergency status

h. Determine the appropriate zone

 acc to category & vehicle type

i. Check available capacity

j. If enough space is available, assign vehicle

k. Otherwise apply applicable alternative zone rule

l. If no suitable zone is available, reject vehicle

m. update occupied spaces & counters

n. Display result

6. Display final parking statistics

7. End

pseudocode

start

Input N

$A = 20$

$B = 40$

$C = 15$

Cars = 0

bikes = 0

vans = 0

rejected = 0

For $i = 1$ to N

Repeat

Input vehicle Type

UNTIL vehicle Type = C OR

vehicle Type = B or vehicle Type = V

Repeat

Input category

UNTIL category = F OR

category = S or category = G

Repeat

Input permit

until permit = Y or permit = N

If permit = N Then

Repeat

Input emergency

until emergency = Y or
emergency = N

If emergency = N Then
display "rejected"

Invalid permit

rejected = rejected + 1

Continue

END If

END If

assigned = FALSE

If category = F then

If AS = 1 then

Display "Assigned To

Room A"

$A = A - 1$
assigned: True
Else

~~also~~ display "Rejected: Zone
A is full"
End If

Else If category = 5 Then

If vehicle type = v Then

If $c \geq 2$ Then
display "Van
redirected to zone c"

$C = C - 2$

assigned: True

Else

Display "Rejected:
No space in zone c"
End If

Else

If $B \geq 1$ Then
display "Assigned To
Zone B"

$B = B - 1$

assigned = True

Else

Display "Rejected:

Zone B is full"

End If

End If

Else If category = C then

If vehicle type = v then

If $C \geq 2$ then

Display "Assigned to
Zone c"

$C = C - 2$

assigned = True

Else

display "Rejected:
No suitable space"

End If

Else

If $C \geq 1$ then

Display "Assigned to
Zone c"

$C = C - 1$

assigned = True

Else

Display "Rejected: Zone c is
full"

End If

End If

End If

If assigned = True then

If vehicle type = c then

Cars = Cars + 1

Else If vehicle type = B then

bikes = bikes + 1

Else

Vans = Vans + 1

End If

Display "Remaining zone
capacity"

Else

rejected = rejected + 1

End If

End for

Display "parked cars" = cars

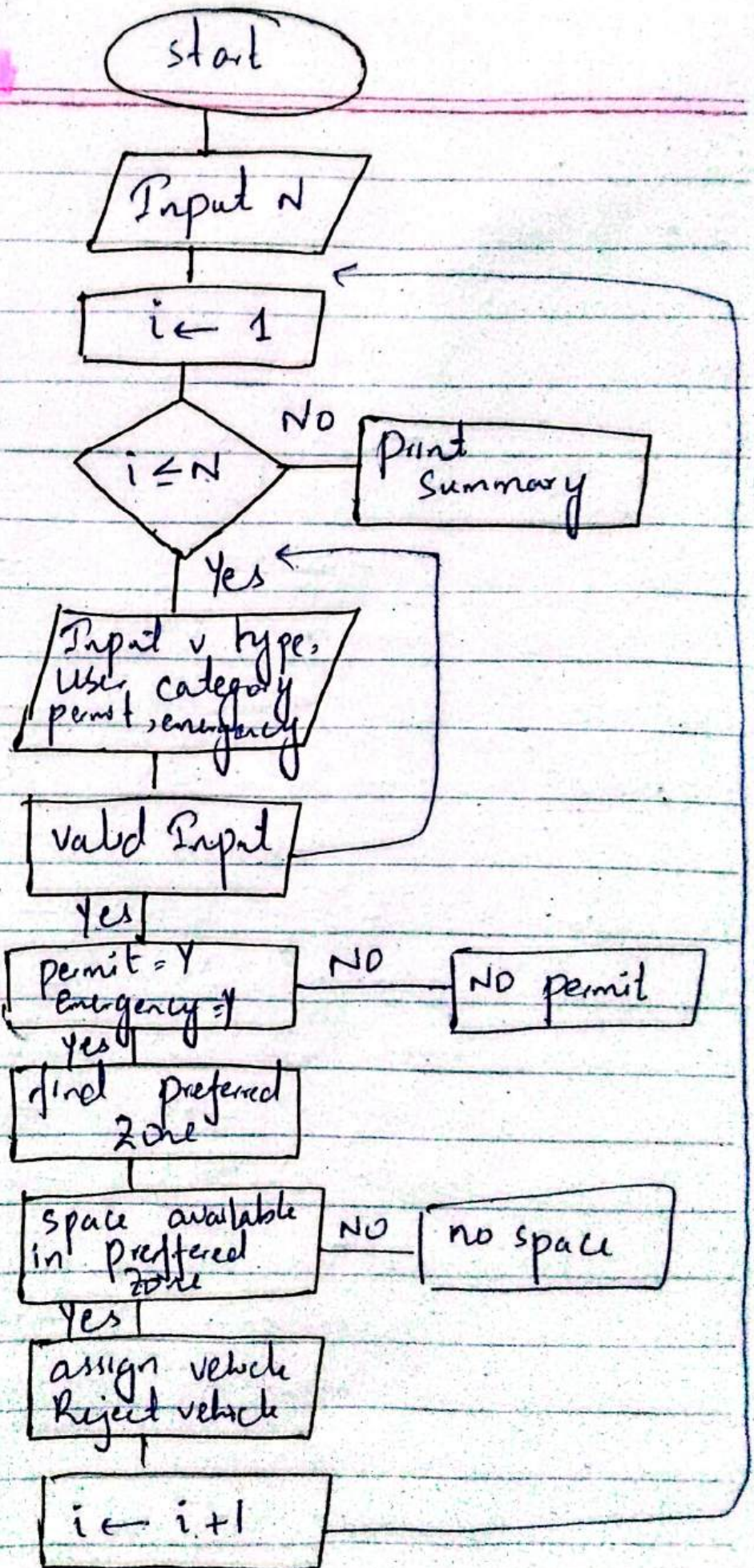
Display "parked bike" = bikes

Display "parked vans" = vans

Display "Rejected vehicle =",
rejected

END

flow chart



Question #6:

IPD#

Input

v-type,
battery level
req. level,
emp hours,
current time,
member, disabled-
priority,
Station-available

process

→ validate
Station
availability
→ check EV/
hybrid
→ calculate
required
charging
assign priority
→ calculate
charging cost
with on peak
off peak
membership
discount
→ calculate
parking cost
→ check (on
stay warning)

output

→ vehicle
Type
→ current
battery
→ Required
battery
→ priority
→ Time Status
→ charging
cost
→ parking cost
→ final payable
amount
→ warning
message.

PAC chart

Problem

→ Decide EV admission charging priority

→ calculate charging parking cost with discount and display summary.

Process

→ check station
→ check eligibility
→ calculate req-charge
→ Assign priority
→ calculate cost
→ Apply discounts
→ Generate warning

Input

→ v-type
→ Battery level
→ req level
→ exp hours
→ current time
→ membership
→ disabled-priority
→ station available

output

→ vehicle type
→ Battery %
→ request charge percentage
→ priority
→ peak/peak-off
→ charging cost
→ Total warning message

Pseudocode:

Start

Input v.type, Battery level, req level,
exp hours, current time, membership,
disabled priority, station available

If station not available

If hybrid then parking only else
no slot AND stop

If hybrid OR battery ≥ 40 then
No EV charging

req charge = req level - battery level

If recharge ≤ 0 then "NO charging
required"

If battery ≤ 15 and req ≥ 80 then
Priority = emergency

Else if disable = Y OR membership = Y
AND battery ≤ 30 then priority = priority

Else priority = Normal

Calculate charging cost based on
peak/off peak and membership.

Calculate parking cost based on
hours and discounts

If hours > 8 then Long-stay warning
else standard

Display all results + total amount

END

Algorithm +

1. Start

2. Input vehicle type, battery level, required level, exp hours, current time.

3. Input membership, disabled priority station available.

4. If Station available = N then

 If vehicle type = H then

 Print charging unavailable - parking only

 Else

 Print No charging slot available

 Stop

End If

End If

5. If vehicle Type = H then

 Print vehicle does not qualify for

 EV charging

 Charging Allowed = false

Else if v type = E and battery level ≥ 40

then

print vehicle doesnot qualify for
EV

charging allowed: false

Else

charging allowed: true

END If

Req charge = req level - battery level

If req charge ≤ 0 then

print NO charging required

charging allowed = false

END If

If charging allowed = True then

If battery level ≤ 15 and req
level = 80 then

Priority = emergency charging priority

Else if disable priority = y or

membership = y AND battery level ≤ 30

then

Priority = Normal charging

END If

END If

If priority $\neq$ emergency charging priority

AND membership = y then

rate = rate * 0.8

END If

If charging allowed = true then

charging cost = req charge * rate

Else

charging cost = 0

END If

If exp hours ≤ 2 then parking cost = 200

Else If exp hours ≤ 5 then parking cost = 400

Else parking cost = 700

If disabled priority = y then parking cost = 0

Else If membership = y then parking cost =

parking cost * 0.8

If exp hours > 8 then

warning = long-stay warning:

Else

warning = standard parking duration

END If

Total = charging cost + parking cost

print "vehicle type", type

print "current battery", battery level, %

print "priority", priority

print "time status", time type

Print "charging cost: Rs", charging cost
 Print "parking cost, Rs", parking cost
 Print "final payable amount; Rs", total
 print warning

Stop

flow chart

